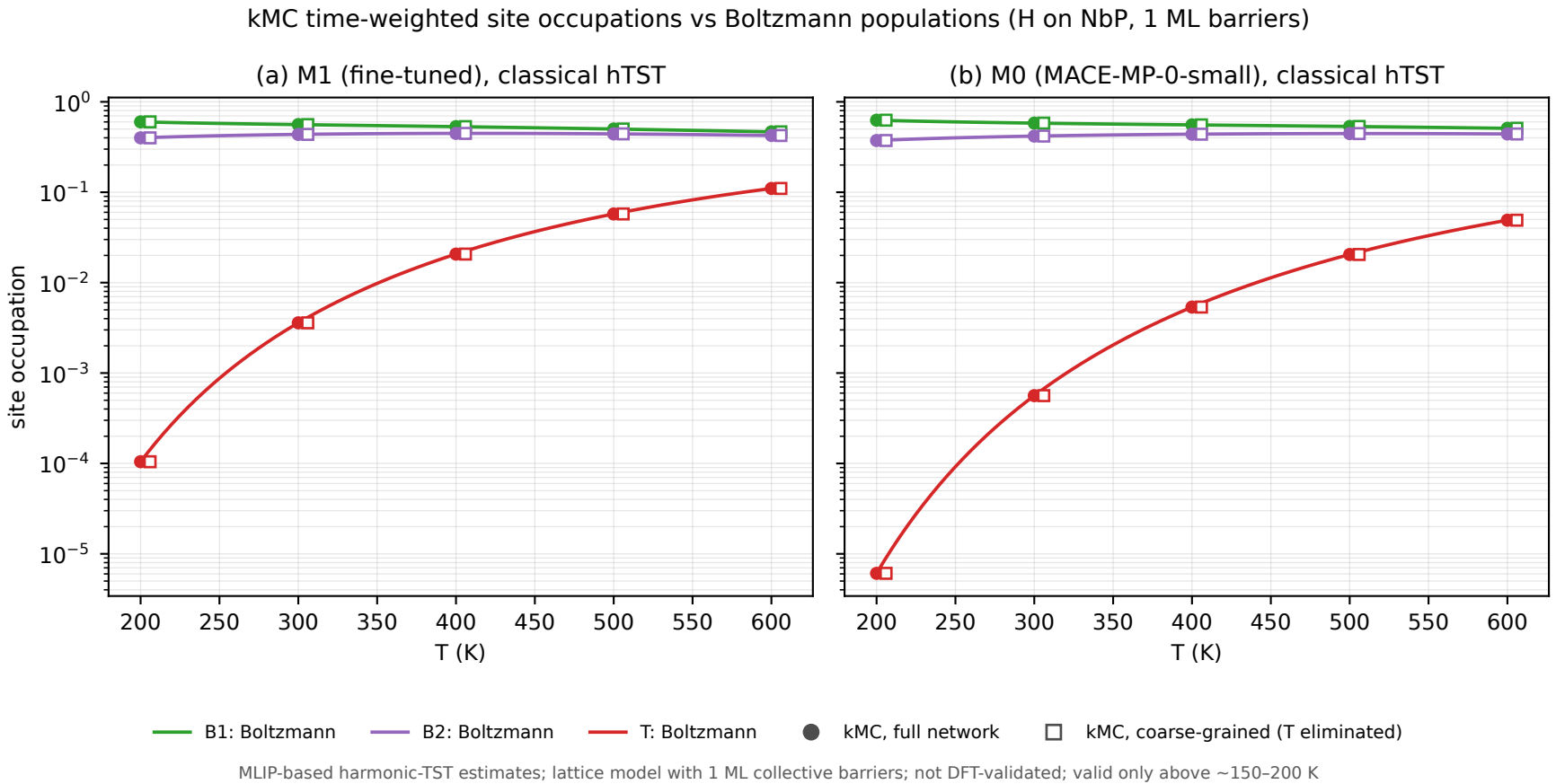
kMC time-weighted site occupations vs Boltzmann populations (H on NbP, 1 ML barriers)
(a) M1 (fine-tuned), classical hTST
(b) M0 (MACE-MP-0-small), classical hTST
site occupation
T (K)
B1: Boltzmann
B2: Boltzmann
T: Boltzmann
kMC, full network
kMC, coarse-grained (T eliminated)
MLIP-based harmonic-TST estimates; lattice model with 1 ML collective barriers; not DFT-validated; valid only above ~150–200 K